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From the Hazards of Death to the Hazards of Indication Creep for TAVR

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On behalf of INTEGRITTY – International Evidence Grading Research Initiative Targeting Transparency and Data Quality

**Equal contribution status.*

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Dear Editor,

We read with great interest the recent meta-analysis by Reddy et al.(1) published in *The Journal of American College of Cardiology* (Volume 85) comparing transcatheter aortic valve replacement (TAVR) and surgical aortic valve replacement (SAVR) in patients described as "lower risk." While the pooled analysis suggests a lower hazard of death with TAVR at 5 years, we believe that key methodological considerations limit the generalizability of these findings.

First, the trials included in the meta-analysis are inhomogeneous in terms of follow-up duration. Only two of the six trials (NOTION and PARTNER 3), representing less than 25% of the total sample (1,280 of 5,397 patients), reported outcomes beyond 5 years. The remaining trials contributed data with shorter follow-up periods (≤1 year), potentially amplifying early advantages of TAVR, while underrepresenting long-term risks such as structural valve deterioration or conduction disturbances. Importantly, the recently published 5-year results of the Evolut Low Risk trial(2), not included in the meta-analysis, showed a mortality trajectory that increasingly favored SAVR over time.

Second, the categorization of patients as "lower risk" is inconsistent with the commonly accepted STS-PROM <4% threshold. UK-TAVI and DEDICATE, in particular, enrolled patients with low-to-intermediate risk profiles. When we replicated the meta-analysis using reconstructed individual patient data (3), inclusion of all trials yielded results similar to Reddy et al., with a significant advantage for TAVR. However, when UK-TAVI and DEDICATE were excluded, the difference in mortality disappeared (p=0.4), suggesting that risk misclassification and trial selection may influence the findings (Figure 1).

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This underscores the importance of consistent definitions and long-term follow-up. The use of a global hazard ratio without assessment of proportional hazards or time-varying treatment effects further complicates interpretation. In rapidly evolving fields such as TAVR, interim data should be interpreted with caution to avoid procedural “indication creep”, the inappropriate expansion of a therapy to patient groups not adequately represented in the evidence base.

In conclusion, while Reddy et al. contribute meaningfully to the ongoing debate on TAVR vs SAVR, we advocate for rigorous risk stratification and longer-term data to support durable and safe therapeutic decisions, especially in younger and lower-risk populations. In the context of TAVR, where enthusiasm for minimally invasive approaches is often exuberant, attention to the accuracy of clinical messaging and adherence to evidence-based indications are essential to maintain the integrity of scientific communication directed to both clinicians and patients to optimize patient outcomes, and support responsible healthcare resource utilization(4).

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Figure Legends

Figure 1. Kaplan-Meier plots for all-cause death comparing TAVR vs. SAVR. **A.** Including low-risk trials and two low-intermediate risk trials (UK TAVI, DEDICATE). **B.** Including low-risk trials only.

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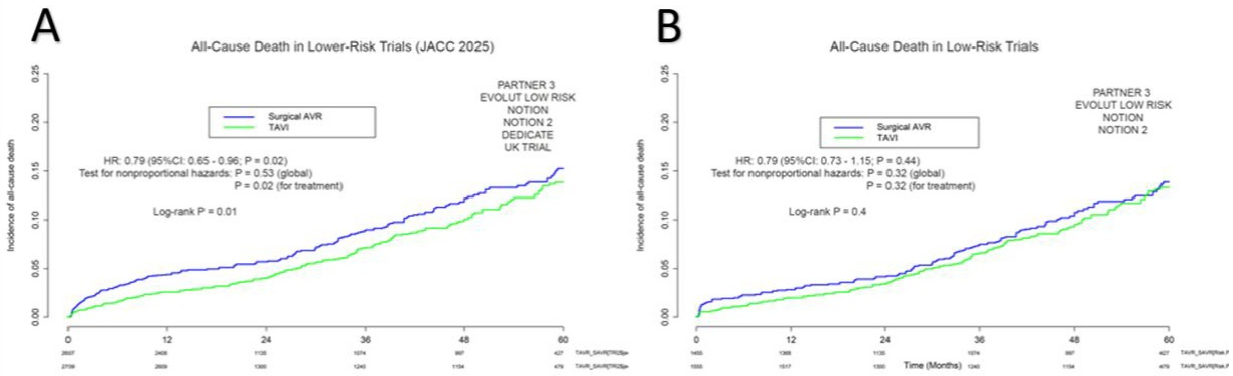

Figure 1: Figure 1

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